

# Bias · Fairness · Accountability

Apple Card offered lower limits to women with identical profiles (2019). Amazon's AI penalised "women's" on resumes (2018). An Indian police face-rec system had 94% false-positive rate on dark-skinned suspects (2022). Each cost its company brand + money + regulatory scrutiny.

Which of these systems would you have...

L1

## WHAT'S INSIDE

- 01 The 5 pillars
- 02 Three fairness metrics
- 03 5 sources of bias
- 04 10-minute fairness audit
- 05 Analogy

WHY THIS LESSON MATTERS

# Apple Card. Amazon hiring. Indian policing. Three tens-of-crores mistakes.

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WHY

## Real

A real reason this matters in industry today.

SCALE

## Today

A real reason this matters in industry today.

SPEED

## Now

A real reason this matters in industry today.

IMPACT

## Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Name the 5 pillars of AI ethics.

2

Define 3 fairness metrics.

3

Identify sources of bias.

4

Run a fairness audit in 10 minutes.

5

Write an ethics impact statement.

AGENDA

# What we'll cover today

Short, sequenced parts. Each one builds on the last.

**01**

## The 5 pillars

Every production AI has a named owner per pillar

**02**

## Three fairness metrics

IMPOSSIBILITY: you cannot satisfy all three unless base rates are equal. Pick one — document why.

**03**

## 5 sources of bias

Know which yours has

**04**

## 10-minute fairness audit

fairlearn + aif360 automate it

# 01

PART 1 OF 4

## The 5 pillars

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Every production AI has a named owner per pillar

CONCEPT 1 OF 4

# The 5 pillars

Fairness — similar inputs, similar outputs across groups.

Accountability — someone owns the failures.

Transparency — users + regulators understand the system.

Privacy — personal data protected throughout pipeline.

Safety — no harm through misuse or failure.

OWN IT

5 pillars

Diffuse ownership = no ownership.

DEFINITION

## The 5 pillars

Every production AI has a named owner per pillar

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EXAMPLE

*Real-world example: applies directly to Indian industry contexts.*

# Three fairness metrics

IMPOSSIBILITY: you cannot satisfy all three unless base rates are equal. Pick one — document why.

## KEY POINTS

- IMPOSSIBILITY: you cannot satisfy all three unless base rates are equal. Pick one — document why
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

CONCEPT 3 OF 4

# 5 sources of bias

Historical — the world that made the data is biased.

Representation — groups under-sampled.

Measurement — features correlate with group membership.

Aggregation — one model for diverse populations.

Deployment — used in ways designers didn't intend.

DIAGNOSE

Source before fix

Mis-diagnosed bias → wrong mitigation.

DEFINITION

## 5 sources of bias

Know which yours has

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EXAMPLE

*Real-world example: applies directly to Indian industry contexts.*

CONCEPT 4 OF 4

# 10-minute fairness audit

1. Identify protected attributes.
2. Compute outcome rates per group.
3. Confusion matrices per group.
4. Pick fairness metric; compute.
5. Document findings + mitigations.

## TOOLS

fairlearn + aif360

From Microsoft + IBM; open-source.

## KEY POINTS

- 1. Identify protected attributes
- 2. Compute outcome rates per group
- 3. Confusion matrices per group
- 4. Pick fairness metric; compute

# 02

PART 2 OF 4

## Engage and reflect

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Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •  
SHARE

ACTIVITY

# Pause and reflect

## PROMPT

*Apply the four concepts above to one real example from your own week.*

## HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

# Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

**Q1**

*Where will you apply this in your next project?*

**Q2**

*What was the single biggest 'aha' from this lesson?*

**Q3**

*Which habit will you start using this week?*

# 03

PART 3 OF 4

## Knowledge check

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Three short questions. One minute each. Honest answers only.

## Can all three fairness metrics be satisfied?

A

Always

B

Never unless base rates equal

C

Only in small models

D

Yes

*Can all three fairness metrics be satisfied?*



CORRECT ANSWER

Never unless base rates equal

WHY

Mathematically impossible with unequal base rates

## Top Indian fairness libraries:

A

Only sklearn

B

fairlearn + aif360

C

Only TensorFlow

D

Only PyTorch

*Top Indian fairness libraries:*

CORRECT ANSWER



**fairlearn + aif360**

WHY

From Microsoft + IBM

## Historical bias means:

A

Random

B

World that made the data was biased

C

Training bug

D

DB error

*Historical bias means:*



CORRECT ANSWER

**World that made the data was biased**

WHY

Data reflects biased historical reality

# 04

PART 4 OF 4

## Apply and close

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One thing to remember. One thing to ship this week.

SUMMARY

# Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

## WORKED EXAMPLE

# How this lesson plays out in one real Indian context

## THE SCENARIO

Fairness — similar inputs, similar outputs across groups.  
Accountability — someone owns the failures.  
Transparency — users + regulators understand the system.  
Privacy — personal data protected throughout pipeline.  
Safety — no harm through misuse or failure.  
OWN IT  
5 pillars  
Diffuse ownership = no ownership.

## THE TAKEAWAY

*The right tool, applied to the right problem, beats the loudest idea every time.*

## WHAT TO NOTICE

- 1 The 5 pillars
- 2 Three fairness metrics
- 3 5 sources of bias
- 4 10-minute fairness audit

# Audits are the new normal.

"Every production AI will be audited. Build the audit in from day one."

## DELIVERABLES

- NEXT
- Lesson 2 — AI Governance.
- Document one decision you made because of this lesson.
- Document one decision you made because of this lesson.

## WHAT 'GOOD' LOOKS LIKE

ONE THING TO REMEMBER

“

*Knowing the right tool is half the battle. Doing the small thing today is the other half.*

— AI Learning Hub

END OF LESSON 8

# What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.